

Mbeya University of Science and Technology

Routing Protocols

A.H.Mbaga
ayubumbaga@gmail.com

Routing

- Process of moving information from a source to a destination across the internetwork.
- Routing takes place at Layer 3 (the network layer) of the OSI model.
- Typically, networks employ a combination of static and dynamic routing.
 - Static routing is preferable for small networks,
 - dynamic routing is ideal for large networks.

Routing Protocols

- Routing protocols are mechanisms for exchanging routing information between routers to make routing decisions.
- Routing protocols can facilitate effective and efficient communication between computer networks.
- these protocols facilitate the secure delivery of data to its destination.

Routing Protocols categorization

■ Behavior:

- Classful (legacy)
- classless protocol.

■ Purpose:

- Interior Gateway Protocol (IGP)
- Exterior Gateway Protocol (EGP)

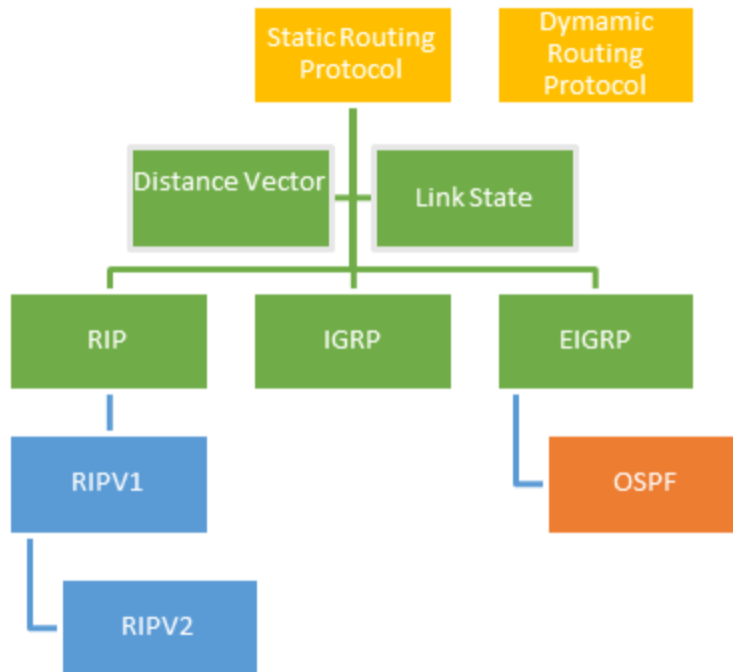
■ Operation:

- Path-vector protocol,
- distance vector protocol
- link-state protocol.

IPv4 Routing protocols categorization

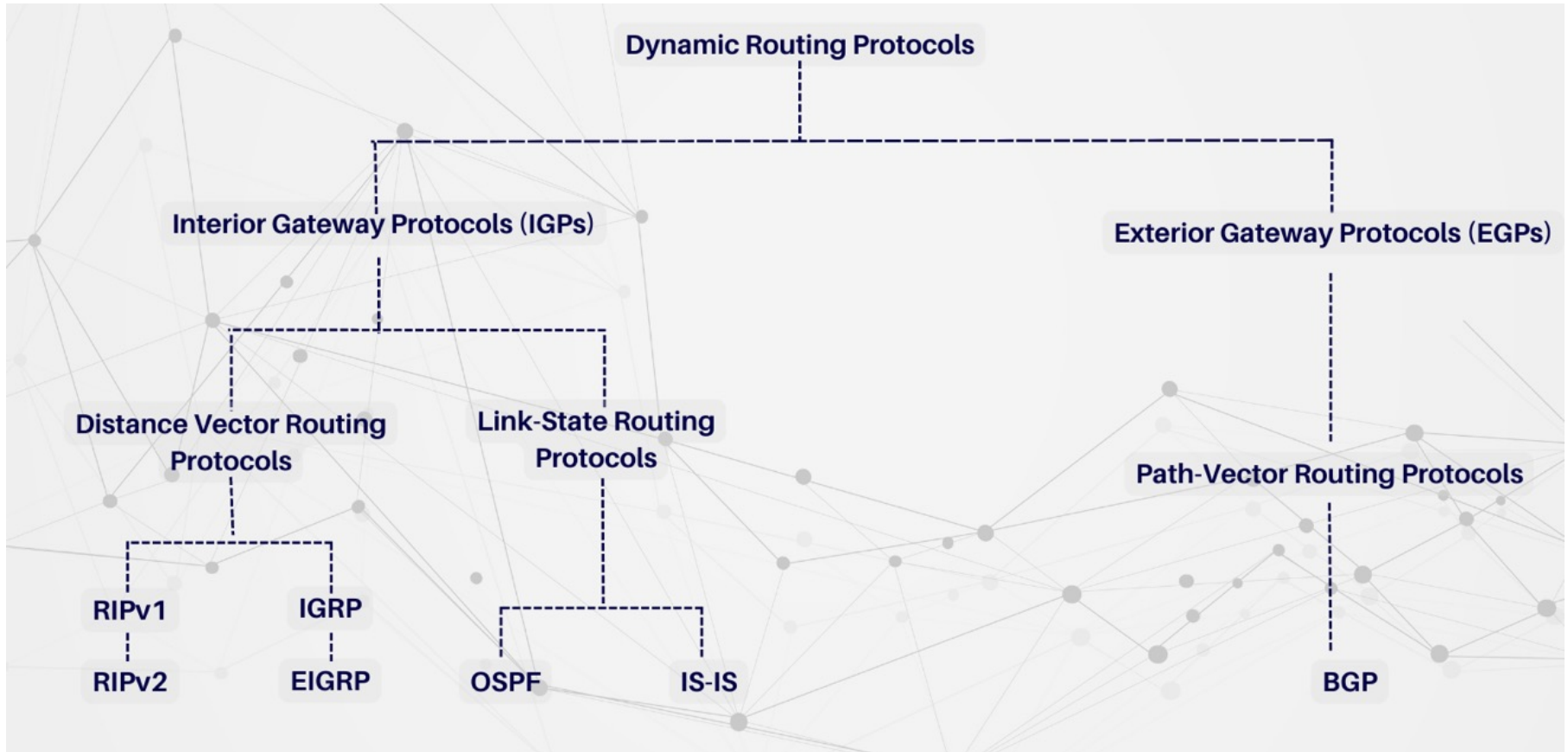
- **RIPv1 (legacy):** IGP, distance vector, classful protocol
- **RIPv2:** IGP, distance vector, classless protocol
- **OSPF:** IGP, link-state, classless protocol
- **IGRP:** IGRP (legacy) is Cisco's IGP, distance vector, classy protocol
- **EIGRP:** IGP, distance vector, classless protocol
- **EGP**
- **BGP:** EGP, classless path-vector protocol

Routing Protocols



Routing Protocols

Dynamic Routing Protocols



Static Routing Protocols

- Static routing protocols are used when an administrator manually assigns the path from source to the destination network.
- It offers more security to the network.
- Advantages
 - Only the administrator is able to add routes
 - No overhead on router CPU.
 - No unused bandwidth between links.

Disadvantages

- The administrator must know how each router is connected.
- Not an ideal option for large networks as it is time intensive
- Whenever link fails all the network goes down which is not feasible in small networks

Dynamic Routing Protocols

- Routers add information to their routing tables from connected routers automatically
 - send out topology updates whenever the network changes' topological structure.
- Advantages
 - Easier to configure even on larger networks
 - It will be dynamically able to choose a different route in case if a link goes down.
 - It helps you to do load balancing between multiple links

Dynamic Routing Protocols -2

■ Disadvantages

- Updates are shared between routers, so it consumes bandwidth
- Routing protocols put an additional load on router CPU or RAM

Principle components of dynamic routing protocols

- *Algorithm:* Routing protocols utilize algorithms to facilitate routing information and to determine the optimal path
- *Routing Protocol Messages:* Messages are used by routing protocols to discover neighboring routers, communicate routing information, and conduct other network-related duties,
 - learning and maintaining accurate network information.
- *Data Structures:* Routing protocols generally utilize tables or databases to perform their tasks. This data is maintained in RAM

Interior gateway (IGP) and exterior gateway (EGP) protocols

- Interior gateway protocols usually exchange information with other routers within single autonomous systems
- include one routing network or a group of networks operating under the same control
- allows for simpler information changes within the same internal network, without the interference of outside routers.

Cont..

- Exterior gateway protocols communicate information between routing networks in different autonomous systems
- they're in different systems, this process is more complex and potentially less common

Classful and classless protocols

- Classful protocols perform routing updates without including subnet mask information
- classification might exclude the information because it focuses on identifying entire networks rather than individual IP address identification.
- Classful protocols require greater bandwidth because they perform routing updates at regular intervals.

Classless protocols

- subnet mask information during routing updates, and they can communicate with devices in separate networks.
- focus on transferring more detailed data at higher complexity levels.
- subnet mask information might help produce more accurate routes for data efficiency
- use less bandwidth because they perform updates only when there are changes to the data.

Routing information protocol(RIP)

- First defined in RFC 1058 as a first-generation routing protocol for IPv4.
- Distance-vector routing protocol that uses the metric hop count.
- Straightforward to configure,
 - making it an excellent option for small networks.
- RIP updates are contained in a UDP segment with both the source and destination ports set to UDP port 520.

Routing information protocol(RIP) -2

- RIPv1 possesses the following qualities:
 - The number of hops is utilized as the path selection metric
 - Every 30 seconds, routing updates are transmitted
 - Greater than 15 hops is considered infinite (too far).
- RIPv2 brought the subsequent enhancements:
 - Security: It includes an authentication mechanism for securing routing table update communications between neighbours.
 - Classless routing protocol support:
 - Improved efficiency.
 - Reduced routing entries: Manual route summarization on any interface is supported.

Open Shortest Path First(OSPF)

- The most prevalent link-state routing protocol
- OSPF development began in 1987, and there are currently two active versions:
 - OSPFv2: OSPF for IPv4 networks
 - OSPFv3: OSPFv3 is the IPv6 version of OSPF
- Implements the link state routing algorithm and is utilized in medium- to large-sized networks.
- Intradomain routing protocol that only operates within a specific routing domain
- Also a hierarchical routing protocol that may be used in a single autonomous system.

OSPF -2

- Enables multipath routing and uses one or more routing metrics
 - dependability, bandwidth, latency, load, and maximum transmission unit (MTU)
- a link-state, interior gateway, and classless protocol that uses the shortest path first (SPF) algorithm to ensure efficient data transmission.
- This type maintains numerous databases containing topology tables and network-wide information.

OSPF-3

- utilizes the Dijkstra algorithm to recalculate paths when topology changes occur
- also employs authentication procedures to maintain the security of its data throughout network modifications and intrusions
- Due to its scalability, OSPF may be advantageous for both small and large network enterprises.

Interior Gateway Routing Protocol (IGRP)

- a distance vector protocol
- it employs several routing metrics (not just hop count) to compute the destination's distance
 - Hold-downs, split horizons, and poison-reverse updates are IGRP features aimed at improving network stability.
- should only be utilized if your current environment consists solely of IGRP and you do not wish to add another routing protocol.

Interior Gateway Routing Protocol (IGRP) -2

- It utilizes many of RIP's fundamental functionalities but increases the maximum number of hops supported to 100.
- compares network parameters such as capacity, dependability, and load to function.
- automatically updates when modifications, such as route modifications, occur.

IGRP protocols routing goals

- The capacity to manage many "types of services" with a single set of data
- Routing loop prevention
- Routing stability, even in extremely large or complex networks
- Low overhead, indicating that IGRP should not consume more bandwidth than it needs for its own operation
- Rapid reaction to varying network structure
- Split traffic along parallel routes when their desirability is equal.
- Consideration of error rates and traffic levels on various paths

IGRP measures

- **Bandwidth of the path section with the smallest bandwidth.**
- **Topological delay time.**
- **Dependability of the route**
 - path's reliability based on the number of packets that really arrive at the destination, relative to the total number of packets transmitted.
- **Path occupancy of the channel.**
 - the percentage of bandwidth currently in use. This value will fluctuate frequently as network traffic fluctuates.

Parameters evaluated by IGRP

- *Hold-downs:* Used to prevent a regular update message from re-establishing a route that may have previously become invalid.
- *Split horizons:* Used to prevent routing loops from occurring between two routers.
- *Poison-reverse updates:* Used to reduce loops between many routers. When the metric rises dramatically,

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- *Update Timer:* The update timer specifies how frequently update messages are transmitted. *Invalid Timer:* The invalid timer specifies how long a router will wait before declaring a route invalid if it is not receiving routing update messages.
- *Hold-time Period:* will indicate the duration of the hold-down period.
- *Flush Timer:* The flush timer specifies the amount of time that must elapse before a route is removed from a routing database.
- *Sleep Timer:* The sleep timer specifies how long update messages will be delayed.

Enhanced Interior Gateway Routing Protocol (EIGRP)

- a distance vector routing protocol used in IP, AppleTalk, and NetWare networks.
- a proprietary Cisco protocol that was developed to succeed the earlier IGRP protocol in 1992
- enhances productivity, lowers routing changes, and facilitates secure message exchange.

Features of EIGRP

- **Rapid convergence**
- **Bounded triggered updates**
- **Management of the topology table**
 - Maintains in a topology table all routes received from neighbour's, not only the optimal ones
- **Hello keepalive mechanism**
 - periodic exchange of a short Hello message is used to maintain adjacencies between routers.
- **Multiple network layer protocol support**
 - supports protocols other than IPv4 and IPv6, including legacy IPX and AppleTalk, because it employs Protocol Dependent Modules (PDM).

Exterior Gateway Protocol(EGP)

- was a routing protocol used to connect autonomous systems on the Internet from the mid-1980s to the mid-1990s when it was replaced by the Border Gateway Protocol (BGP).
- EGP Protocol routing table includes:
 - Network addresses of nearby devices
 - Route costs
 - Identified routers

Exterior Gateway Protocol(EGP)

- maintains network databases close to one another to route the various paths data may travel to reach its destination.
 - The databases then distribute the information to the connected routers so that all routers' tables are current
 - The updated routing tables can assist in determining the optimal data route.
 - the devices can update their routing tables and select network paths based on more accurate information.

Border Gateway Protocol(BGP)

- an alternative exterior gateway protocol that was created to replace EGP
- employs the optimal path selection technique for data package transfers, making it a distance vector protocol.
- To automatically find the optimal route, BGP refers to the following variables:
 - Adjacent IP addresses
 - Router designation
 - Path distance
 - Origin type

Border Gateway Protocol(BGP)- 2

- Best Path Selection Algorithm is utilized to determine the optimal paths for data packet transfers
- Will select routes with the shortest path to the destination.
 - If no special parameters have been configured
- Enables administrators to modify transfer routes based on:
 - their requirements and provides extensive security measures to ensure that only authorized routers can exchange data and information.

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- The algorithm for selecting the optimal route path can be modified by modifying the BGP cost community attribute.
- Able to make routing decisions based on factors including :
 - weight, local preference, locally generated, AS Path length, origin type, multi-exit discriminator, eBGP over iBGP, IGP metric, router ID, cluster list, and neighbor IP address.
- only transmits updated routing table data when a change occurs
- BGP protocol can be verified so that only authorized routers can exchange data.

Intermediate System-to-Intermediate System(IS-IS)

- Designed by (ISO)
- IS-IS was originally built for the OSI protocol suite and not TCP/IP.
- Was formerly known as the routing protocol used mostly by ISPs and carriers, although enterprise networks are increasingly adopting it
- Employs a modified form of the Dijkstra algorithm
 - the protocol groups routers together to build bigger domains and connects routers for data transport

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- Employs these two network types frequently:
 - **Network Service Access Point (NSAP):** Similar to an IP address, a network service access point (NSAP) identifies a service access point in systems that employ the open system interconnection (OSI) concept.
 - **Network Entity Title(NET):** This facilitates the identification of specific network routers within bigger computer networks

Comparison of Routing Protocols

- **Implementation and maintenance**
- **Speed of convergence**
- **Classful vs Classless**
- **Scalability**
- **Resource usage**

Summary

- A rapid convergence
- Easy to configure
- Permits optimal route selection
- Minimize update traffic
- Provides for loop-free routing
- Adapts to alterations
- Supports vary in length
- Compatible with established hosts and routers
- Scales to a substantial size